

INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

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Project Report

Module 3 - AutoSenti: AI-Based Automotive Review and Customer Sentiment Analytics

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Abstract

AutoSenti is an AI system designed for analyzing customer reviews of automobiles through aspect-based sentiment analysis. It focuses on automotive topics such as engine, battery, mileage, safety, comfort, service, infotainment, and price, predicting whether sentiments regarding each topic are Positive, Negative, or Neutral. The system utilizes a configurable keyword-based aspect extractor for identifying relevant aspects and employs an aspect-conditioned transformer classifier based on BERT or XLM-RoBERTa for sentiment inference. Results are displayed via an interactive Streamlit dashboard, which includes confidence scores and probability breakdowns. This project offers a practical, explainable method for transforming unstructured automotive feedback into actionable insights that can aid in product improvement, customer-service evaluation, and informed purchasing decisions.

1. Introduction

Online automotive reviews contain valuable information about customer experiences, opinions, and perceptions of different vehicle features. Customers frequently discuss aspects such as engine performance, mileage, battery, safety, comfort, infotainment, service quality, and price. However, manually reading and organizing a large number of reviews is time-consuming and inefficient. As the volume of online reviews increases, extracting meaningful information from them becomes increasingly difficult.

A major challenge is that a single review may mention multiple vehicle aspects while expressing different sentiments toward each one. For example, a customer may describe the mileage as excellent but criticize the infotainment system. Traditional sentiment analysis generally identifies only the overall sentiment of a review and does not clearly determine which specific vehicle feature the sentiment refers to.

AutoSenti addresses this limitation using Aspect-Based Sentiment Analysis (ABSA). The system accepts a written automotive review, identifies the relevant aspects mentioned in the text, and classifies the sentiment associated with each aspect as positive, negative, or neutral. The application uses transformer-based Natural Language Processing techniques to understand the contextual meaning of review text. AutoSenti is designed as an academic and research-oriented dashboard that presents aspect-level sentiment results through a simple and user-friendly interface, helping users understand detailed customer feedback more efficiently.

2. Concept and Working Principle

The central concept is aspect-level sentiment analysis. An aspect is a specific feature or service discussed in a review, while sentiment represents the opinion expressed about that aspect. For example, in the review "The mileage is excellent but the service is expensive," the system should identify Mileage as Positive and Service as Negative.

The system works through the following sequence:

1. The user enters an automotive review in the Streamlit interface.
2. The input is validated and cleaned by the preprocessing pipeline.
3. The aspect extractor compares normalized review text with the configured automotive keyword map.

4. Each detected aspect is paired with the complete review.
5. The selected transformer model evaluates the aspect-review pair.
6. Softmax probabilities are converted into Positive, Negative, or Neutral predictions.
7. The dashboard displays the detected aspects, sentiment labels, confidence scores, and probability details.

The aspect-conditioned input uses the aspect as Sentence A and the complete review as Sentence B. This gives the classifier the context needed to estimate sentiment toward a particular aspect rather than only predicting the overall sentiment of the review.

3. Project Overview

AutoSenti is a focused web application developed for analyzing and interpreting automotive customer reviews using Natural Language Processing and transformer-based models. The system identifies vehicle-related aspects mentioned in a review and predicts the sentiment associated with each individual aspect. This enables users to understand not only the overall opinion expressed in a review but also the specific strengths and weaknesses associated with different vehicle features.

The current product is implemented as a single-page **Streamlit dashboard** designed for quick and interactive review analysis. Users can enter an automotive review into the application, select an available transformer-based model, and initiate the analysis. The system processes the input text, detects relevant automotive aspects, and generates sentiment predictions for each identified aspect. The results are presented along with confidence scores, providing an indication of the model's prediction certainty.

By presenting aspect-level results in a structured and accessible interface, AutoSenti makes transformer-based sentiment analysis easier to understand and use. The application is primarily intended for academic experimentation, research, and demonstration of aspect-based sentiment analysis in the automotive domain.

4. Dataset

The models were fine-tuned on the **MuseASTE** dataset, an aspect-based sentiment analysis benchmark derived from the **SemEval** shared tasks on opinion mining. MuseASTE provides structured (aspect, opinion, sentiment) triplets aligned to review sentences, which were adapted to the (aspect, review, sentiment) sequence-pair format required by this project.

Property	Value
Dataset name	MuseASTE (SemEval-based)
Task format	Aspect-review sentence pair classification
Sentiment classes	Positive, Negative, Neutral
Access	Downloaded automatically at training time via the data adapter
Source domain	Multi-domain (restaurants, laptops) - adapted for automotive fine-tuning

10.2 Data Split

The dataset was split into three subsets:

Split	Rows	Purpose
Train	6,222	Model parameter updates
Test (internal)	1,099	Held-out evaluation during training
Devel / Validation	2,998	Hyperparameter guidance and final reporting
Total	10,319	-

The train/test split was created by reserving 15% of the original training data as an internal test set. The devel set was kept separate throughout training.

10.3 Class Distribution (Training Split)

Sentiment	Count	Share
Positive	4,205	67.6%
Negative	1,174	18.9%
Neutral	843	13.5%
Total	6,222	100%

5. Objectives

- Detect automotive aspects in natural-language customer reviews.
- Classify sentiment for each detected aspect as Positive, Negative, or Neutral.
- Support multiple aspects in one review.
- Present predictions in a clear, professional dashboard.
- Keep model selection and inference simple for end users.
- Provide confidence values to support interpretation of model predictions.

6. Scope

Included

- Single vehicle-review analysis.
- Keyword-based automotive aspect extraction.

- Aspect-conditioned transformer sentiment classification.
- BERT and XLM-RoBERTa model options.
- Confidence scores and probability breakdowns.
- Confidence overview visualization.
- Responsive Streamlit user interface.

Excluded from the current product

- Batch review upload through the dashboard.
- Dataset management and annotation tools.
- Model training and evaluation screens.
- Export workflows.
- The original MuseASTE research dataset and related experimental code.

7. Supported Aspects

The default taxonomy contains nine automotive aspects:

- Vehicle
- Engine
- Battery
- Mileage
- Safety
- Comfort
- Service
- Infotainment
- Price

Aspect names and keyword mappings are configured in `config.yaml`, allowing the taxonomy to be extended without changing the extraction code.

8. System Architecture



Main layers

Layer	Responsibility
app/	Streamlit page, layout, theme, controls, and result components
src/data/	Review cleaning and input validation
src/inference/	Aspect extraction, sentiment classification, and orchestration
src/models/	Tokenizer/model loading and sentiment label mappings
src/analytics/	Confidence chart generation
config.yaml	Aspect taxonomy, sentiment labels, and model definitions

9. Methodology

9.1 Input validation

The application checks that the review is a non-empty text string. Very short reviews are rejected, while unusually long reviews generate a warning before preprocessing.

9.2 Text preprocessing

The preprocessing pipeline:

1. Removes HTML tags and URLs.
2. Removes control characters.
3. Normalizes Unicode text.
4. Expands common English contractions.
5. Collapses repeated whitespace.
6. Truncates text to the configured maximum length.

9.3 Aspect extraction

Aspect extraction is keyword-based. The review is normalized and compared with the configured keyword map. Every matching aspect is passed to the sentiment classifier.

This approach is transparent and fast, but it does not detect implicit aspects that are not represented by configured keywords.

9.4 Sentiment classification

For every detected aspect, the system creates an aspect-review pair:

- Sentence A: the aspect name.
- Sentence B: the complete review.

The pair is passed to a sequence-classification transformer. The model returns probabilities for Positive, Negative, and Neutral sentiment. The highest-probability class becomes the displayed prediction.

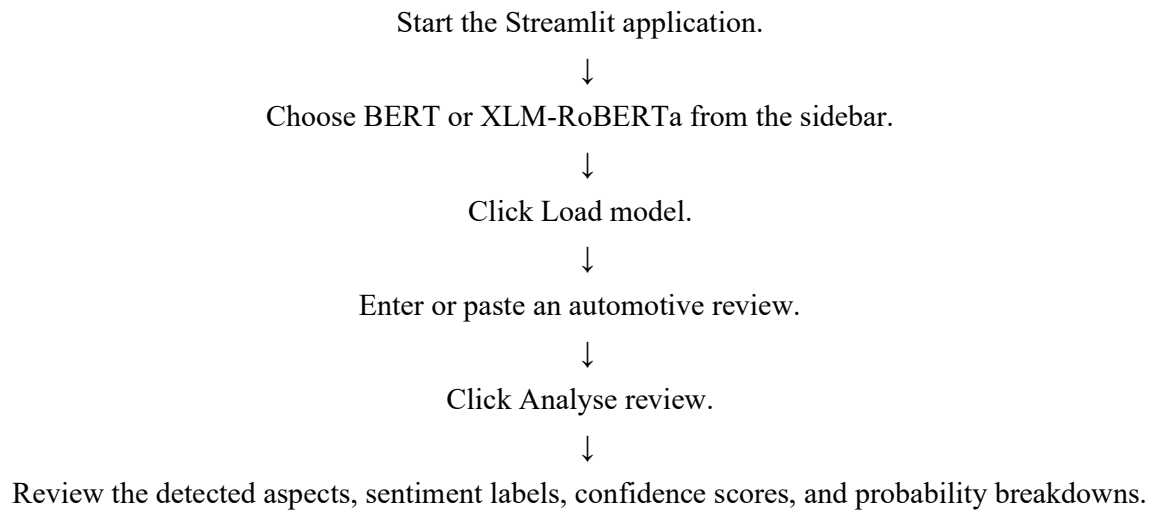
10. Models

Model	Hugging Face identifier	Intended use
BERT	<code>bert-base-uncased</code>	English reviews and efficient inference
XLM-RoBERTa	<code>xlm-roberta-base</code>	Multilingual reviews and broader generalization

The application first checks for a local fine-tuned checkpoint. If no checkpoint is available, it loads the configured pretrained model from Hugging Face.

Local checkpoints are excluded from Git through `.gitignore` because they are large generated artifacts.

10. User Workflow



11. User Interface

The dashboard uses a single-page layout with:

- A compact model-control sidebar.
- A focused review input area.
- Example review buttons.
- A summary row for detected aspects and sentiment counts.
- Individual aspect sentiment cards.
- Confidence visualization and raw result details.

The visual system uses a light slate background, white surfaces, teal primary actions, and restrained positive, negative, neutral, warning, and error colors.

12. Experimental Results

12.1 BERT - Overall Metrics

Split	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
Validation / Devel	74.28%	63.28%	61.33%	62.21%
Test	77.07%	68.16%	63.91%	65.74%

The model achieves **77.07% test accuracy** after 5 epochs of fine-tuning on 6,222 training samples.

12.2 BERT - Per-Class Metrics (Validation Split - 2,998 samples)

Class	Precision	Recall	F1-Score	Support
Positive	0.8251	0.8603	0.8424	2,062
Negative	0.5926	0.5594	0.5755	429
Neutral	0.4808	0.4201	0.4484	507
Macro avg	0.6328	0.6133	0.6221	2,998
Weighted avg	0.7336	0.7428	0.7376	2,998

12.3 BERT - Per-Class Metrics (Test Split - 1,099 samples)

Class	Precision	Recall	F1-Score	Support
Positive	0.8258	0.8869	0.8553	743
Negative	0.6949	0.5942	0.6406	207
Neutral	0.5242	0.4362	0.4762	149
Macro avg	0.6816	0.6391	0.6574	1,099
Weighted avg	0.7603	0.7707	0.7635	1,099

12.4 BERT - Confusion Matrix

Validation Split (2,998 samples)

	Pred: Positive	Pred: Negative	Pred: Neutral
True: Positive	1,774	109	179
True: Negative	138	240	51
True: Neutral	238	56	213

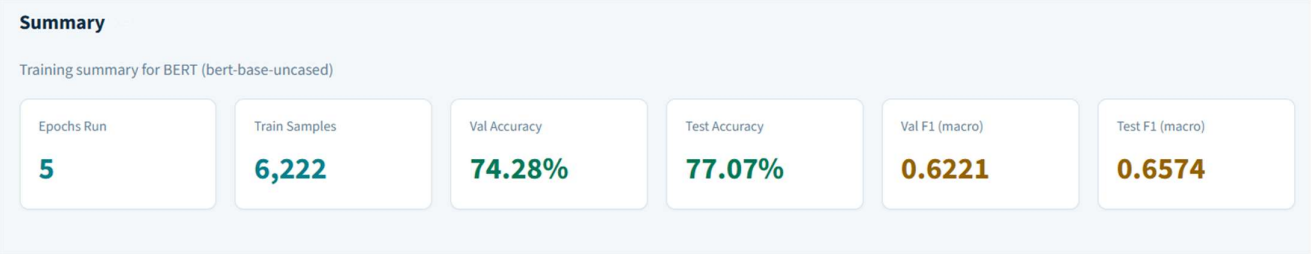


Fig: BERT Model Summary



Fig: BERT Accuracy Graph



Fig: BERT Confusion Matrix

Split	Class	Precision	Recall	F1-Score	Support
Validation	Positive	0.8251	0.8603	0.8424	2,062
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Fig: BERT Class Metrics Table

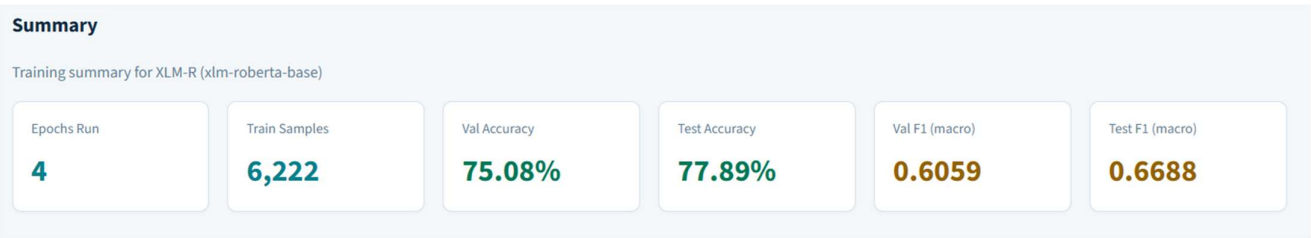


Fig: XLM-RoBERTa Model Summary

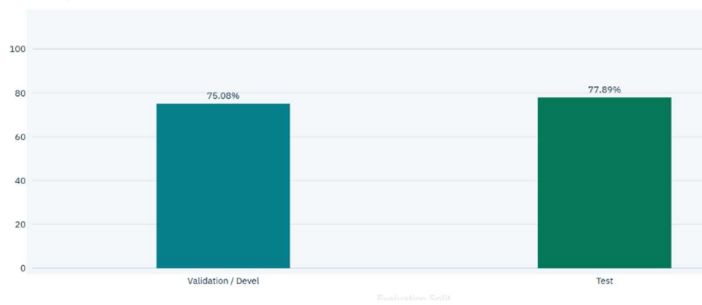


Fig: XLM-RoBERTa Accuracy Graph

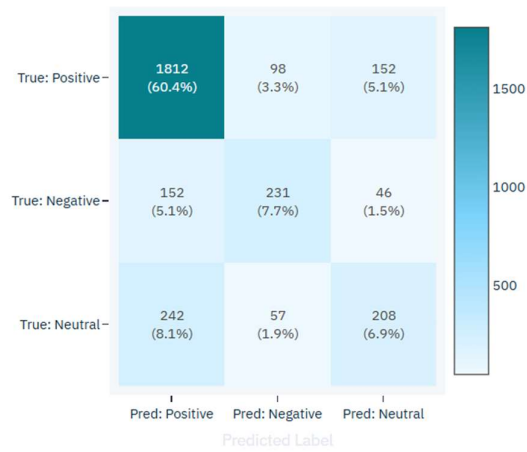


Fig: XLM-RoBERTa Confusion Matrix

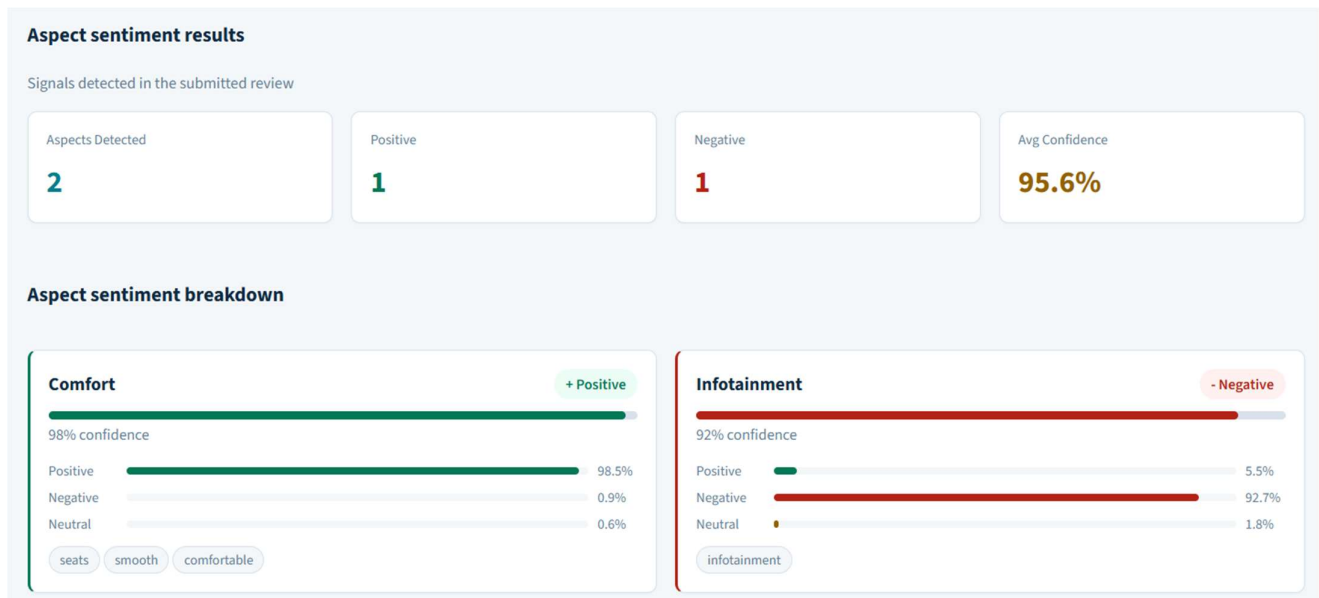


Fig: Model Output

12. Limitations

- Aspect extraction depends on configured keywords.
- Implicit or indirectly described aspects may not be detected.
- Sarcasm, irony, and figurative language may reduce prediction quality.
- Confidence values are model probabilities, not guarantees of correctness.
- Pretrained fallback models may not provide task-specific accuracy until fine-tuned.
- Model loading can be slow on CPU and requires memory for transformer weights.

13. Future Improvements

- Add a controlled batch-analysis workflow if bulk review processing becomes necessary.
- Replace or supplement keyword extraction with a trained aspect-extraction model.
- Add domain-specific evaluation metrics and a versioned test set.
- Add multilingual review examples and language detection.
- Provide model version and checkpoint metadata in the interface.
- Add automated UI tests for the primary review-analysis workflow.

14. Conclusion and Future Scope

AutoSenti demonstrates how natural language processing and transformer models can be combined to analyze automotive customer feedback at the aspect level. The current system provides a complete workflow from review input and preprocessing to aspect detection, sentiment classification, confidence visualization, and result presentation. Its configurable taxonomy and support for both BERT and XLM-RoBERTa make it suitable for experimentation with English and multilingual automotive reviews.

The main limitation is that aspect extraction currently depends on predefined keywords. Future work can improve the system by introducing a trained neural aspect-extraction model, expanding multilingual support, adding batch review analysis, and evaluating performance on a labeled domain-specific benchmark. Other useful extensions include explainable predictions, model and checkpoint versioning, review export, human feedback for error correction, and monitoring of sentiment trends across vehicle models or time periods.

15. References

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